

Alexander Peczon

Mission



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BakedSoups



apeczon

Full stack software engineer specializing in backend databases, frontend platforms, and microservice automations. Experienced building data driven systems that ship to real users at production scale.

Education

University of San Francisco

Graduated 2026

B.S. Computer Science

Relevant Coursework: Linear Algebra, Statistics, Vector Calculus, Software Engineering, Computer Architecture, Algorithms

Experience

Future Tilt - Software Engineer (Part-time)

Jul 2025 – Present

Built full stack applications, data pipelines, and workflow automation systems

San Francisco

- Developed a React and FastAPI platform that converted campaign plans into editable Klaviyo compatible marketing emails, using AI generated copy and reusable email components to produce branded first drafts from designer maintained assets in Google Drive, reducing email preparation time by 40%.
- Engineered a campaign synchronization service using AWS Lambda, BigQuery, and webhook based APIs to coordinate campaign data across Trello, Klaviyo, and internal systems, reducing campaign maintenance overhead by 50%.
- Collaborated with a Data Consultant to write SQL transformations and ETL pipelines that unified Shopify transaction and Klaviyo engagement data into BigQuery, enabling advanced customer segmentation and campaign targeting across 15+ ecommerce brands, datasets exceeding 20M+ records, and processing 100K+ daily insertions.

Alaris Security - Junior Full Stack Engineer

Aug 2025 – Dec 2025

Developed cybersecurity automation and compliance systems

San Francisco

- Consolidated 50+ frontend database queries into typed tRPC procedures backed by Drizzle ORM, improving maintainability and reducing redundant data access across a Next.js + React monorepo.
- Developed a cybersecurity analysis system with a team of 6 that analyzed millions of security records across Elastic, CrowdStrike and Microsoft Defender API, to identify malicious activity and generate customer specific NIS2 compliance reports, reducing report generation time by 30%.
- Resolved frontend and data consistency issues across an internal React platform, eliminating duplicate data displays, fixing UI rendering bugs, and establishing reusable patterns for frontend to backend communication.

MAGICS Lab - NLP Research Assistant

Mar 2025 – May 2026

Built datasets and models for explainable sentiment analysis

San Francisco

- Built parallelized ETL pipelines in Go across 5 virtual machines using DuckDB analytical querying to generate entity aware NLP datasets from 20,000+ news articles.
- Implemented a modular explainable ABSA framework that outperformed VADER on SemEval 2014 Restaurant by 5 percentage points (78.58% vs 73.57% accuracy), with transparent reasoning traces.

University of San Francisco - Data & Web Engineering Intern

Jul 2024 – Jul 2025

Marketing Analytics & Enrollment Insights

San Francisco

- Conducted year-over-year exploratory analysis on 500,000+ prospect student records from SLATE, identifying macro enrollment trends analyzing acceptance and declination rates by geography, finding that recruitment events where students from the same region met had significantly higher conversion rates.
- Automated recurring website content updates for prospective students by building Python and Jinja2 templating workflows over a legacy SLATE-hosted system running raw HTML and React, eliminating manual copy paste updates across enrollment marketing pages.

Projects

NextSteamGame.com Recommendation Engine (Open Source)

Feb 2025 - May 2025

Explainable Similarity Search & Recommendation Engine

San Francisco

- Built an explainable recommendation system for 80,000+ Steam games that decomposed game identity into weighted semantic profiles, allowing users to customize recommendations around attributes such as soundtrack, setting, gameplay systems, and narrative themes.
- Developed a pipeline that analyzed up to 2,000 reviews per game, applying heuristic filtering, quality scoring, and semantic classification to generate recommendation features, semantic vectors, and niche identity tags.
- Designed a retrieval architecture that precomputed recommendation candidates offline and applied user controlled reranking at runtime, enabling fast recommendations for 80,000+ games on a resource constrained cloud deployment.
- Grew the platform to 30,000+ users, with recommendation interactions driving discovery across 8,000+ unique games.

USF Search Engine Crawler

2024

Concurrent Web Crawler and Search Engine in Go

San Francisco

- Built a concurrent web crawler and search engine in Go that downloaded and indexed the USF website in approximately 10 seconds, compared to roughly 30 minutes for sequential crawler implementations.
- Designed a worker pool architecture with 300 extraction workers using configurable job queues and batched processing to concurrently fetch, parse, and store website content in SQLite for searchable retrieval.

Technical Skills

Languages: Python, Go, SQL, TypeScript, JavaScript

Frameworks & Libraries: React, FastAPI, PyTorch, Pandas, Scikit-learn

Infrastructure & Data: PostgreSQL, BigQuery, Docker, AWS Lambda, Airflow, DuckDB, ChromaDB

Cloud & Tools: AWS, Google Cloud, Linux, Git, Nginx